

Session 121: 5G/6G Signal Processing and ISAC

TIME 2:30 PM – 4:00 PM IST VENUE SP 1.20 (TVS Lab)



- 121.1** **Low Complexity FPGA Implementation of 2x2 MIMO-OTSM Transceiver for 6G Communications**
Sapta Girish Neelam
- 121.2** **Bayesian Multi-Snapshot Learning for Sparse Doubly-Selective CSI Estimation in DCO-OTFS VLC Systems**
Shubham Saxena, Suraj Srivastava, Aditya K. Jagannatham
- 121.3** **Joint Recovery of Two Carriers from One-bit Sampled Signals**
Aparna Agrawal, Kumar Appaiah, Sibi Raj B Pillai
- 121.4** **Robust URLLC Link Adaptation via Statistical Modeling of Time-Varying SINR Deviations**
Shubham Baranwal, Ugrasen Singh, Manav Bhatnagar
- 121.5** **Real-Time Ephemeris Update Mechanism for Effective Doppler and Timing Pre-Compensation in 5G NR-NTN gNB**
Akula Chinna Veeresh
- 121.6** **DMRS-Driven Deep Learning Network for Doppler Estimation in 5G NR-NTN**
Ipsita Sannigrahi, Sathish Nallamolu, Pradeep Goutam, Naru Pradeep
- 121.7** **Differential Cross-Attention based Network for Robust Modulation Classification in OTFS Systems**
Tonmoy Rajkhowa, Sanjeev Sharma, Kuntal Deka, Tharmalingam Ratnarajah
- 121.8** **RCoD-AoANet: Residual Covariance Denoising for AoA Estimation in ISAC-OTFS Vehicular Systems**
Vidyarenu Swamy, Kanwardeep Singh Gahlot, Sandeep Joshi
- 121.9** **Learning-Based Delay, Doppler, and Angle Estimation Using Quantized DDAE for OFDM Bistatic ISAC**
Shiv Shankar, Radhakrishna Ganti, Klutto Milleth J
- 121.10** **On the Joint Impact of UAV Orientation Instability and Phase Errors in UAV-Aided RIS Systems**
Sourosmi Das, Manojkumar B. Kokare, Prashant Sharma, Swaminathan R., Sumit Gautam, Saptarshi Ghosh
- 121.11** **SRS-Based CSI Quality Assessment and Selection for MU-MIMO Precoding in 5G NR**
Lalit Bharti, Adarsh Ravi, Hamza Bouchebbah, Krishnan Venkataraghavan, Seshashayi Thalluri, Preetam Kumar
- 121.12** **Dual-Stream CNN--LSTM Fusion for Pilot-Assisted Automatic Modulation Classification in OTFS Systems**
Dhruvil Mali, Kamal Captain, Dayshaun kakadiya, Arvind Chandak
- 121.13** **Cross-Generation TDD Sensing for 6G ISAC: A Feasibility Study with Legacy LTE and NR Infrastructure**
Anindya Saha
- 121.14** **SNF-PRP: A Covert Integrating Sensing and Communications Framework**
Dhruvil Bhatt, Vidushi Kumar
- 121.15** **Low-Latency Deep Learning Based Doppler Velocity Estimation for Integrated Sensing and Communication**
Samarth Sharma Bhardwaj, Aakanksha Tewari, Syed Asrar Ul Haq, Sumit J Darak

Session 122: Multimodal Learning and Prediction

TIME 2:30 PM – 4:00 PM IST VENUE SP Building Side Foyer



122.1 AgentBook: Enabling Fine-Grained Editability in Visual Narratives via Page-Centric Agent Coordination

Ayushman Sarkar, Zhenyu Yu

122.2 Spectral Decoupling in Attention Heads: Tracking the Onset of Memorization in Vision Transformers

Dibyajyoti Jena, Sumantra Dutta Roy, Vinayak Abrol

122.3 Do Latent Representations of Deep Visual Architectures Follow the Population Manifold Hypothesis of the Primate AIT Cortex?

Ankit Sharma, Sumohana Channappayya

122.4 AVT-PAC: A Pipeline For Multimodal Action Prediction and Captioning

Athira Krishnan R, Ambarish Parthasarathy, Sucharitha Devarakonda, Lakshmi Sai Rithika Srikrumam, Swapna M, JV Satyanarayana, Sumohana Channappayya

122.5 Character-aware Subtitling through Improved Alignment and Diverse Priors

Adhiraj Deshmukh, Makarand Tapaswi, Vineet Gandhi

122.6 Task-Adaptive FFN Editing for Continual Blind Image Quality Assessment

Satish Maurya, Parimala Kancharla

122.7 EchoVision: A Decoupled Audio-Visual Retrieval Framework for Video Question Answering

Subhajit Sarkar, Joyita Chakraborty, Sagnik Nandi, Suman Samui, Subrata Nandi

122.8 Towards an Efficient and Unified Strategy for Video Understanding Applications

Rahul Kumar, Sumohana Channappayya

122.9 Bidirectional Convolutional-GRU Integrated GAN for Unpaired Low-Light Image Enhancement

Palak Deb Patra, Santosh Panda, Manoj Kumar Bishwal, Pankaj Kumar Sa

122.10 Getting Uncertainty Right: Local Calibration for Wind Power Forecasting

Manush Sanchela, Vrishin Rajpurohit, Pritam Anand

122.11 Microstate temporal dynamics predict valence but not arousal in tactile stimulation

Suranjita Ganguly, Mohan Raghavan, Kousik Sarathy Sridharan

122.12 VGGT-Based COLMAP-Free Initialization for Sparse-View 3D Gaussian Splatting

Shreeya Venkatraman, Yuvan Raj Krishna, Jiji C V

Session 123: Biomedical Vision and Sensing

TIME 2:30 PM – 4:00 PM IST VENUE ECE 1.07



123.1 A Slice-Based Deep Network for Brain Age Prediction From MRI

Amisha Rath, Sanjay Ghosh

123.2 cAAE-EchoGAN: A Conditional Adversarial Autoencoder Framework for Joint Echocardiographic Image and Segmentation Mask Generation

Shivani Singh, Ashutosh Singh, Shanti Chandra

123.3 Scale-aware Economy-Accuracy Trade-off in Contactless Palmprint Recognition Using 14-Tap Q-Shift DTCWT and Layer-Wise Attention

Divyang Jadav, Siddhant Dongare, Inderjeet Inderjeet, Saket Pateriya, Deekshant Kumar, Hemant Goklani, Vikram Gadre

123.4 DFANet: Dual-Branch Feature Fusion with Multi-Attention Decoding for Colonoscopy Polyp Segmentation

Srinija Kakkanti, Harshini Mamidi, Srikanth Panigrahi

123.5 Swin-DSA-Net: A Dual-Stage Attention Guided Swin Transformer Network with Multi-scale Refinement for Skin Lesion Segmentation

Nitin Yadav, Bibek Goswami, Marjana Bhuyan, M.K. Bhuyan

123.6 A 3D Channel-Guided Layer Attention Network for Layer-Specific Structural Biomarker Analysis in Optical Coherence Tomography

Mane Pooja, Manish Bhatt, Samarendra Dandapat

123.7 SmartVitals-BP: Contactless Blood Pressure Estimation from Facial Videos via Multi-Scale Temporal Modeling and Metadata Fusion

Srinivas Routhu, Aradhana Mishra, Anoushka Panigrahi, Sanjay Swain

123.8 SwinNetraGF: A Gated Attention Fusion Transformer for Retinal Disease Classification

Anish Kumar Vishwakarma, Preity Preity, Rakesh Ranjan

123.9 A Multiscale Wavelet-Graph Spectral Approach for Cardiovascular Risk Assessment Using Retinal Fundus Images

S Shri Vatssan, Malaya Kumar Nath